

Towards predictive maintenance: A performance evaluation framework for cooling towers in HVAC systems

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ABSTRACT

Cooling towers are essential for heat rejection in heating, ventilation, and air conditioning (HVAC) systems, especially for large scale buildings, but have historically received less research attention compared with other HVAC equipment. However, with the increasing adoption of “multi-tower – multi-pump – multi-chiller” configurations and the widespread integration of variable frequency drives (VFDs) in cooling towers and condenser water pumps for the purpose of energy saving, the demand for the operational optimization and predictive maintenance of cooling towers has grown significantly. To address this need, this study proposes a performance evaluation framework toward predictive maintenance, integrating both physics-informed and data-driven approaches. The framework enables in situ thermal performance assessment and early detection of potential degradation using operational data, without requiring system shutdowns. Maintenance decisions are guided by two key components: (1) characteristic curves derived from Merkel theory, serving as a benchmark for performance evaluation, and (2) model predictive accuracy and prediction interval (PI) reliability metrics, which indicate performance degradation and potential maintenance benefits. The proposed framework was validated using real-world operational data from a data center cooling plant. Compared to existing evaluation methods, it eliminates the reliance on complex lookup tables, interpolation, or intrusive testing. By providing a scalable and practical solution, the framework supports energy-efficient and reliable cooling tower operation, and serves as a foundation for predictive maintenance deployment.

1. Introduction

Cooling towers are critical components in cooling systems, as their heat exchange capabilities have a direct impact on chiller efficiency and the reliability of the entire system [1,2]. However, due to their lower energy consumption compared to chiller units, cooling towers often receive less focus in the development of operational and maintenance strategies [3–5]. In recent years, a growing number of cooling systems have been equipped with variable frequency drives (VFDs) for cooling towers and condenser water pumps, enabling more flexible energy-saving control strategies [6–8]. Additionally, the traditional configuration of cooling systems, characterized by a “one tower – one pump – one chiller” setup, is increasingly being replaced by “multi-tower – multi-pump – multi-chiller” arrangements, where any tower and pump can serve any chiller [9]. This flexible configuration allows for systematic equipment rotation, facilitating scheduled maintenance and expanding the system’s energy-saving potential [10]. During operation,

higher-efficiency units can be prioritized, while maintenance is performed on less efficient ones as needed, ensuring the cooling system maintains a high level of efficiency and reliability over extended periods [11,12]. This shift encourages new considerations for evaluating cooling tower thermal performance, aiming to develop optimal cooling tower sequencing strategies and maintenance schedules based on the real-time heat exchange efficiency of each tower in the system.

Currently, maintenance of cooling systems primarily relies on preventive (time-based) and corrective (failure-based) maintenance strategies. However, with advancements in internet of things (IoT) technology and computational resources, predictive (condition-based) maintenance, which enables proactive maintenance of HVAC systems by detecting and preventing faults, has gained increasing attention [13]. By continuously collecting real-time operational data and employing physical or empirical models to predict expected equipment performance, deviations between predicted and actual operating conditions can be identified. When significant discrepancies emerge, they indicate potential performance degradation, allowing timely intervention and

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Nomenclature		
Abbreviations		
HVAC	Heating, ventilation, and air conditioning	
VFD	Variable frequency drive	
IoT	Internet of things	
PI	Prediction interval	
RMSE	Root mean square error	
SST	Sum of squared total	
SSE	Sum of squared error	
RMSE	Root mean square error	
CVRMSE	Coefficient of variation of the RMSE	
MSE	Mean squared error	
PICP	PI coverage probability	
MPIW	Mean prediction interval width	
NMPIW	Normalized MPIW	
PINC	PI nominal confidence	
CWC	Coverage width-based criterion	
BMS	Building management system	
CTI	Cooling technology institute	
CT	Cooling tower	
CWP	Condenser water pump	
CHI	Chiller	
NTU	Number of heat transfer unit	
Symbols		
$T_{cws}/T_{w,o}$	Condensing water supply temperature, [°C] or [K]	
$T_{cwr}/T_{w,i}$	Condensing water return temperature, [°C] or [K]	
Me_M	Merkel number (Merkel model)	
Me_P	Merkel number (Poppe model)	
Me_e	Merkel number (ϵ – NTU model)	
Le	Lewis factor, $Le = h_C/(c_{pma} h_D)$	
V_{ref}	The rated or designed air volume flow rate of cooling tower fan, [m ³ /s]	
μ	Equal to the PI nominal confidence	
R	The width of target	
h	Specific enthalpy of the air in cooling tower, [J/kg]	
h_{sw}	Specific enthalpy of saturated air at the same cross-sectional cooling water temperature, [J/kg]	
c_{pw}	Specific heat capacity of liquid water at constant pressure, 4180 [J·kg ⁻¹ ·K ⁻¹]	
T_w	Equivalent temperature of the cooling water, [K]	
ω_{sw}	Saturation humidity ratio of air evaluated at the local bulk water temperature, [kg/kg]	
ω	Humidity ratio, [kg water vapor/kg dry air]	
h_v	Specific enthalpy of the vapor, [J/kg]	
h_w	Specific enthalpy of the water, [J/kg]	
$\dot{m}_w$	Water mass flow rate, [kg/s]	
$\dot{m}_a$	Dry air mass flow rate, [kg/s]	
T_{wb}	Out-door wet-bulb temperature, [°C] or [K]	
T_{range}	CT range temperature, [°C] or [K]	
FR_{air}	The air flow rate ratio (actual air flow rate divided by design air flow rate).	
FR_{water}	The water flow rate ratio (actual water flow rate divided by design water flow rate).	
R_{lg}	Liquid-to-gas ratio (FR_{water} / FR_{air})	
$f_1 - f_{27}$	Coefficients in <i>York</i> model	
$c_1 - c_{35}$	Coefficients in <i>CoolTools</i> model	
$\dot{Q}_{CT}$	Heat dissipation of the cooling tower, [kW] or [W]	
$c_{J,1} - c_{J,3}$	Coefficients in Jin's model	
$c_{M,1} - c_{M,2}$	Coefficients in Ma's model	
$p_1 - p_3$	Coefficients in Zhou's model	
h_{amb}	Specific enthalpy of the ambient air, [J/kg]	
$h_{sa}^{w,i}$	Specific enthalpy of the saturated air at the inlet temperature of the cooling water, [J/kg]	
$\dot{v}_a$	Air volume flow rate, [m ³ /s]	
$\dot{v}_w$	Water volume flow rate, [m ³ /s]	
ρ_w	Water density, [kg/m ³]	
h_D	Mass transfer coefficient, [kg·m ⁻² ·s ⁻¹]	
h_C	Heat transfer coefficient, [W·m ⁻² ·K ⁻¹]	
c_{pma}	Specific heat of the moist air, [J·kg ⁻¹ ·K ⁻¹]	
T_{app}	Approach temperature, [°C] or [K]	
R^2	Coefficient of determination	

proactive maintenance to prevent further efficiency loss or system failures [14]. Proactive and targeted maintenance offers new opportunities to enhance energy efficiency and improve system reliability [15]. For cooling towers, maintenance measures mainly include fan inspections and cleaning of the fill. Fan motor failures are often critical; thus, daily inspections are typically conducted, where abnormal vibrations and noise are monitored to detect potential issues. Scaling of fill material or a decline in fan performance may not lead to immediate severe consequences; however, it gradually reduces thermal performance by decreasing heat exchange efficiency or causing airflow instability. The extent of this degradation is often correlated with the tower's cumulative operational time. In systems where cooling tower fans operate synchronously (at the same speed and in the same mode), running a tower with reduced thermal performance alongside a normally functioning tower leads to a mixed outlet water temperature. This mixing causes the pumps and chillers to consume more energy, ultimately reducing the system's overall efficiency [16]. In systems where each cooling tower fan operates independently, a cooling tower with decreased heat exchange efficiency needs to run its fan at higher speeds to achieve the target condensing water supply temperature. This approach significantly increases cooling tower's energy consumption. Therefore, it is important to promptly monitor performance differences among cooling towers within the system, based on which to take targeted proactive maintenance measures. The implementation of this process relies on an accurate and transferable performance evaluation

method for cooling towers.

In predictive maintenance, the reliability of condition monitoring depends heavily on the accuracy of the performance evaluation model [17]. Over the years, researchers have continually developed cooling tower models to evaluate their thermal performance [18]. The evolution has progressed from physical models aimed at simulating system heat transfer characteristics [19], to empirical models designed for optimizing control [20–23], and more recently, to neural network models [24–29]. Throughout this development, researchers have sought a balance between model accuracy and computational efficiency [30]. While many of these models demonstrate good accuracy within their respective datasets, their applicability to real-world systems remains uncertain. One key challenge is that real-world operational data introduce complexities such as sensor accuracy fluctuations and diverse operating conditions [11]. In contrast, most existing models have been validated using data from controlled experiments, limited operating ranges, or manufacturer-provided datasets, which do not fully capture the variability of practical applications as they fail to account for prediction uncertainty. Additionally, different models require different input and output variables, yet the availability of these variables in actual operations varies, further complicating model implementation. Moreover, cooling tower operation in real-world systems often deviates from design conditions, making performance evaluation under off-design conditions particularly challenging. Current methods still rely on strict filtering of operating conditions, complex lookup tables, interpolation

techniques, or even shutdown measurements, all of which hinder implementation [31].

To enable effective predictive maintenance of cooling towers, two critical research gaps must be addressed:

1. Extracting cooling tower characteristic curves for performance evaluation: How can updated characteristic curves be efficiently derived from operational data to assess performance without requiring system shutdown?
2. Selecting an appropriate performance evaluation model for predictive maintenance: Which cooling tower model best supports predictive maintenance, considering both accuracy and sensor availability?

To address these challenges, this study first proposes a methodology for deriving cooling tower characteristic curves from field data based on physical principles, eliminating the need for system shutdown measurements. Second, it examines existing thermal performance models for cooling towers, incorporating prediction uncertainty and evaluating their accuracy, practical applicability, and considerations for optimization-based control applications. Based on the selected models, a performance evaluation framework toward predictive maintenance is developed to assess cooling tower thermal performance while accounting for uncertainties in sensor measurements and model identification. The framework is demonstrated using operational data from a real-world data center cooling plant. In practice, this approach can be integrated with characteristic curve analysis to provide actionable recommendations for daily maintenance, ultimately enhancing system reliability and operational efficiency.

The remainder of this paper is structured as follows: Section 2 outlines the key steps of the proposed performance evaluation workflow and reviews widely used cooling tower thermal performance models. Section 3 demonstrates the implementation of the proposed methodology using operational data from a real-world data center cooling plant. Section 4 evaluates the model performance and highlights the key contributions of this study. Finally, Section 5 concludes the paper by summarizing the findings and their implications.

2. Methodology

This section introduces the workflow of the proposed cooling tower thermal performance evaluation method and elaborates on the methods involved in each step.

2.1. Workflow overview

The overall workflow of the proposed evaluation process in this study involves three main steps (see Fig. 1): model selection and

identification (Section 2.2); typical condition identification (Section 2.3); performance evaluation (Section 2.4).

2.2. Step 1: Model selection and identification

2.2.1. Data preparation

This step involves the pre-processing of data collected from the building management system (BMS). A critical aspect of this process is analyzing common anomalies in the dataset to assess whether these anomalies indicate system faults. Based on this evaluation, a decision is made to either remove the anomalies as irrelevant outliers or retain them as representative data for equipment faults. To illustrate this process, Table 1 provides a summary of key operational data related to cooling towers from historical operation records, along with some common identified data anomalies. The principal variables examined in relation to cooling tower operations include weather information, which influences the air-water heat transfer dynamics; readings from thermometers and flowmeters, which track the heat extraction from the system; and fan operating status data, which provide insights into the airflow rates. The whole process follows the steps described in our previous research [32]. The refined dataset obtained through these procedures will be utilized for further analysis, ensuring that the subsequent stages of the study are built upon a foundation of reliable and relevant data.

Table 1
Summary of data anomalies in cooling tower historical operation records.

Data source	Measured variable	Types of data anomalies
Weather information provided by thermo-hygrometer	Ambient wet-bulb temperature [°C] Ambient dry-bulb temperature [°C] Ambient relative humidity [%]	Missing data, data drift, fluctuation, data staleness
System status provided by thermometers and flowmeters	Condenser water temperature at cooling tower inlet [°C] Condenser water temperature at cooling tower outlet [°C] Condenser water flow rate [L/h]	
Operating status information	ON/OFF status [0 or 1] Fan operating frequency [Hz] Fan power consumption [kW]	Missing data, incorrect status, data staleness

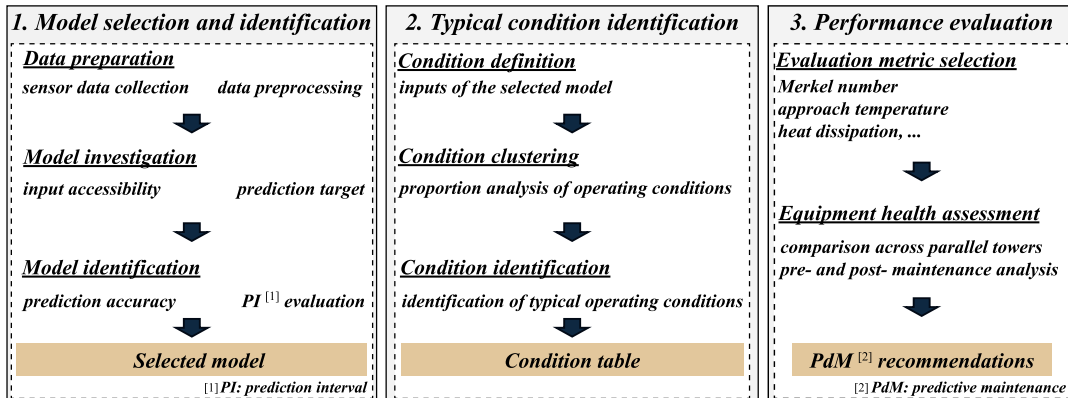


Fig. 1. The implementation process for the proposed cooling tower thermal performance evaluation method.

2.2.2. Investigation of cooling tower thermal performance models

Commonly used models for cooling tower thermal performance prediction (i.e., for predicting outlet water temperature rather than power consumption) can be classified into three categories based on their modeling techniques: physical, hybrid, and empirical. Tables 2 and 3 summarize several representative models along with their associated references, focusing on the available thermal performance indicators, the expressions used in each model, the critical assumptions made during model derivation, and the applicable conditions for each model. These models, each with its unique strengths, are widely utilized in numerical simulations of cooling plants or in the development of advanced control algorithms such as model predictive control.

The Merkel model, the earliest physical-based model, simplifies the heat and mass transfer processes using specific assumptions outlined in Table 3. In contrast, the Poppe model, also physically derived, offers a more detailed formulation of these processes. The ε -NTU model, while adopting similar assumptions to the Merkel model, simplifies the solution process to reduce computational intensity. The purely empirical York and CoolTools models require extensive parameter fitting, involving 27 and 35 fitting coefficients respectively, and their limitations are discussed in prior studies. To enhance real-time prediction efficiency, researchers have developed simplified hybrid models, such as Jin's model and Ma's model. These models employ thermal resistance-based expressions to approximate heat transfer processes and have

demonstrated good predictive accuracy within their datasets. A recent study by Zhou *et al.* introduced a new thermal performance model, validated against various manufacturer datasets, proposing three model variants to predict heat transfer, outlet water temperature, and required airflow. Additionally, although some studies have developed black-box models based on neural networks and system-specific models tailored to unique operating conditions, due to their limited generalizability across different systems, these models are not included in this table.

The complexity of heat transfer processes in cooling towers highlights the need for an appropriate thermal performance prediction model. A comprehensive literature review indicates that the Merkel, Poppe, and ε -NTU models exhibit similar modelling accuracy for outlet water temperature [33]. While the Poppe model provides a more precise estimation of air-side thermal performance, its higher computational complexity limits its practicality. In real-world applications, temperature and humidity sensors are rarely installed at the air outlet, whereas outlet water temperature sensors are more commonly used. Therefore, this study adopts the Merkel model as the baseline physical-based model for evaluating cooling tower thermal performance. Among empirical models, both the York and CoolTools models are utilized in *EnergyPlus*, with the York model also incorporated into the *Modelica Buildings Library* [22]. Due to its more frequent use in prior studies, the York model is selected for parameter fitting in this study. Between Jin's and Ma's models, which have demonstrated comparable predictive performance

Table 2

Summary of widely adopted models for cooling tower thermal performance prediction from the literature.

Ref	Modeling technique	Thermal performance indicator	Presence of coefficients to be fitted [Yes/No]	Expression
1 Merkel [19]	Physical	Me_M	No	$Me_M = \int_{T_{w,o}}^{T_{w,i}} \frac{c_{pw}}{(h_{sw} - h)} dT_w$
2 Poppe [34]	Physical	Me_p	No	$Me_p = \int_{T_{w,o}}^{T_{w,i}} \frac{c_{pw}}{(h_{sw} - h) + (Le - 1)[(h_{sw} - h) - (\omega_{sw} - \omega)h_v] - (\omega_{sw} - \omega)h_w} dT_w$
3 ε -NTU [20]	Hybrid	Me_e	No	$Me_e = \frac{c_{pw}}{dh_{sw}/dT_w} NTU$ when $\dot{m}_a \geq \frac{\dot{m}_w c_{pw}}{(dh_{sw}/dT_w)}$, $Me_e = \frac{\dot{m}_a}{\dot{m}_w} NTU$ when $\dot{m}_a < \frac{\dot{m}_w c_{pw}}{(dh_{sw}/dT_w)}$.
4 York [23]	Empirical	T_{app}	Yes	$T_{app} = f_1 + f_2(T_{wb}) + f_3(T_{wb})^2 + f_4(T_{range}) + f_5(T_{wb})(T_{range}) + f_6(T_{wb})^2(T_{range}) + f_7(T_{range})^2 + f_8(T_{wb})(T_{range})^2 + f_9(T_{wb})^2(T_{range})^2 + f_{10}(R_{lg}) + f_{11}(T_{wb})(R_{lg}) + f_{12}(T_{wb})^2(R_{lg}) + f_{13}(T_{range})(R_{lg}) + f_{14}(T_{wb})(T_{range})(R_{lg}) + f_{15}(T_{wb})^2(T_{range})(R_{lg}) + f_{16}(T_{range})^2(R_{lg}) + f_{17}(T_{wb})(T_{range})^2(R_{lg}) + f_{18}(T_{wb})^2(T_{range})^2(R_{lg}) + f_{19}(R_{lg})^2 + f_{20}(T_{wb})(R_{lg})^2 + f_{21}(T_{wb})^2(R_{lg})^2 + f_{22}(T_{range})(R_{lg})^2 + f_{23}(T_{wb})(T_{range})(R_{lg})^2 + f_{24}(T_{wb})^2(T_{range})(R_{lg})^2 + f_{25}(T_{range})^2(R_{lg})^2 + f_{26}(T_{wb})(T_{range})^2(R_{lg})^2 + f_{27}(T_{wb})^2(T_{range})^2(R_{lg})^2$
5 CoolTools [23]	Empirical	T_{app}	Yes	$T_{app} = c_1 + c_2(FR_{air}) + c_3(FR_{air})^2 + c_4(FR_{air})^3 + c_5(FR_{water}) + c_6(FR_{air})(FR_{water}) + c_7(FR_{air})^2(FR_{water}) + c_8(FR_{water})^2 + c_9(FR_{air})(FR_{water})^2 + c_{10}(FR_{water})^3 + c_{11}(T_{wb}) + c_{12}(FR_{air})(T_{wb}) + c_{13}(FR_{air})^2(T_{wb}) + c_{14}(FR_{water})(T_{wb}) + c_{15}(FR_{air})(FR_{water})(T_{wb}) + c_{16}(FR_{water})^2(T_{wb}) + c_{17}(T_{wb})^2 + c_{18}(FR_{air})(T_{wb})^2 + c_{19}(FR_{water})(T_{wb})^2 + c_{20}(T_{wb})^3 + c_{21}(T_{range}) + c_{22}(FR_{air})(T_{range}) + c_{23}(FR_{air})^2(T_{range}) + c_{24}(FR_{water})(T_{range}) + c_{25}(FR_{air})(FR_{water})(T_{range}) + c_{26}(FR_{water})^2(T_{range}) + c_{27}(T_{wb})(T_{range}) + c_{28}FR_{air}(T_{wb})(T_{range}) + c_{29}(FR_{water})(T_{wb})(T_{range}) + c_{30}(T_{wb})^2(T_{range}) + c_{31}(T_{range})^2 + c_{32}(FR_{air})(T_{range})^2 + c_{33}(FR_{water})(T_{range})^2 + c_{34}(T_{wb})(T_{range})^2 + c_{35}(T_{range})^3$
6 Jin [21]	Hybrid	$\dot{Q}_{CT}$	Yes	$\dot{Q}_{CT} = \frac{C_{J,1} \dot{m}_w^{C_{J,3}}}{1 + C_{J,2} \left(\frac{\dot{m}_w}{\dot{m}_a} \right)^{C_{J,3}}} (T_{w,i} - T_{wb})$
7 Ma [5]	Hybrid	$\dot{Q}_{CT}$	Yes	$\dot{Q}_{CT} = \frac{C_{M,1} \dot{m}_w}{1 + C_{M,2} \left(\frac{\dot{m}_w}{\dot{m}_a} \right)} (T_{w,i} - T_{wb})$
8 Zhou [2]	Hybrid	$\dot{Q}_{CT}$	Yes	$\dot{Q}_{CT} = \frac{p_1 p_3 \dot{v}_a \dot{v}_w (h_{sa}^{w,i} - h_{amb}) (T_{w,i} - T_{wb})}{p_1 \dot{v}_w (T_{w,i} - T_{wb}) + p_3 (\dot{v}_a + p_2 \dot{v}_w) (h_{sa}^{w,i} - h_{amb})}, T_{w,o} = T_{w,i} - \frac{p_1 p_3 \dot{v}_a (h_{sa}^{w,i} - h_{amb}) (T_{w,i} - T_{wb})}{c_{p,w} \rho_w [p_1 \dot{v}_w (T_{w,i} - T_{wb}) + p_3 (\dot{v}_a + p_2 \dot{v}_w) (h_{sa}^{w,i} - h_{amb})]},$ $\dot{v}_a = \frac{c_{p,w} \rho_w (T_{w,i} - T_{w,o}) [p_1 (T_{w,i} - T_{wb}) + p_2 p_3 (h_{sa}^{w,i} - h_{amb})]}{p_3 (h_{sa}^{w,i} - h_{amb}) [p_1 (T_{w,i} - T_{wb}) - c_{p,w} \rho_w (T_{w,i} - T_{w,o})]} \dot{v}_w.$

Table 3
Assumptions and applicability of common cooling tower thermal performance models.

Ref	Iterative requirement	Model assumptions from the literature	Applicability
1 Merkel	Yes	1. The Lewis factor ^[1] is equal to 1. 2. The air exiting the tower is saturated with water vapor and it is characterized only by its enthalpy. 3. The reduction of water flow rate by evaporation is neglected in the energy balance.	-
2 Poppe	Yes	-	-
3 $\epsilon - NTU$	Yes	Same as the assumptions of the Merkel model.	-
4 York	No	-	See Section 16.1.2.3 in [23]
5 CoolTools	No	-	See Section 16.1.2.3 in [23]
6 Jin	No	1. Heat and mass transfer occur only in the direction normal to the flow. 2. The Lewis factor is assumed to be constant. 3. The reduction in water flow rate due to evaporation is neglected in the energy balance. 4. The cross-sectional area of the tower and the temperature distribution throughout the water stream at any cross-section are considered uniform. 5. The saturation enthalpy is assumed to be approximately linear with respect to the wet-bulb temperature.	-
7 Ma	No	Same as the assumptions of the Jin's model.	-
8 Zhou	No	1. Heat and mass transfer between water and air in the cooling tower occurs only in the direction perpendicular to their flow direction. 2. The Lewis factor is equal to 1. 3. Within the operating range of the cooling tower, the specific enthalpy of the saturated air inside the cooling tower is linearly related to the wet-bulb temperature. 4. There is no heat and mass transferred from cooling tower to the surrounding environment.	-

Note: [1]. Lewis factor (Le): $Le = hc / (c_{p,m} h_0)$, the dimensionless ratio of heat transfer coefficient (hc , ($W \cdot m^{-2} \cdot K^{-1}$)) to mass transfer coefficient (h_0 , ($kg \cdot m^{-2} \cdot s^{-1}$)), where ($c_{p,m}$, ($J \cdot kg^{-1} \cdot K^{-1}$)) is the specific heat of the moist air. $Le = 1$ indicates that the sensible heat lost by the air is equal to the latent heat required for water evaporation in the contact area.

in previous research, Jin's model is chosen for fitting due to its greater number of parameters. Additionally, Zhou's model, as a promising new model, is included for evaluation. This study fits the York model, Jin's model, and Zhou's model, selects the model with the highest fitting accuracy for cooling tower thermal performance prediction, and compares the results with those from the Merkel model as the physical-based reference. This approach aims to identify the most suitable model for practical applications while ensuring reliable thermal performance predictions.

2.2.3. Stochastic model identification and prediction interval (PI) evaluation

To effectively predict the cooling tower's thermal performance, it is important to account for uncertainties in the predictions. These uncertainties primarily arise from three sources: model discrepancies, sparse data, and random errors. Model discrepancy refers to the mismatch between a mathematical model and the real physical system, typically caused by modeling assumptions, simplifications, and other factors. Data sparsity refers to the limitation in available data, insufficient coverage, or low measurement frequency, which hinders adequate model calibration. Random errors refer to data fluctuations caused by uncertainties in measurement, environmental conditions, or numerical computations, which are generally difficult to predict and eliminate.

To assess the predictive accuracy of the models, statistical metrics such as coefficient of determination (R^2) and the coefficient of variation of the root mean square error (CVRMSE) are commonly used. The R^2 is defined as:

$$R^2 = 1 - \frac{SSE}{SST}, \quad (1)$$

where SSE is the sum of squared error, SST is the sum of squared total. A higher R^2 value, approaching 1, suggests that the model explains most of the variance in the data, indicating better predictive performance. The $CVRMSE$ is calculated as:

$$CVRMSE = \frac{RMSE}{\bar{y}} \times 100\%, \quad (2)$$

where $\bar{y}$ is the mean of the observed values, and the root mean square error ($RMSE$) is given by:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (3)$$

where y_i is the represents the observed or actual value, $\hat{y}_i$ is the represents the predicted value by the model, n is the total number of observations. The $RMSE$ quantifies the standard deviation of the residuals and measures the model's prediction accuracy; the $CVRMSE$ provides a normalized measure of prediction error relative to the mean observed values. Lower $CVRMSE$ values indicate better model accuracy.

To further quantify prediction uncertainties, prediction intervals (PIs) are integrated into the stochastic model identification process. By incorporating both residual variance and parameter covariance into the prediction bounds, PIs account for uncertainties arising from sensor measurements and data noise, providing a probabilistic buffer against deviations from true system behavior. This is especially valuable in HVAC applications, where real-time data may be affected by noise, drift, or delayed responses. To ensure the effectiveness of PIs, it is crucial to evaluate their quality by balancing reliability and sharpness. The entire process can be summarized as follows:

For new observation $y_{n+1}(\mathbf{x}_{n+1})$ at the predictor value $\mathbf{x}_{n+1}$,

$$y_{n+1}(\mathbf{x}_{n+1}) = f(\mathbf{x}_{n+1}) + \varepsilon_{n+1}, \quad (4)$$

where $f(\mathbf{x}_{n+1})$ is the fitted value when the predictor is $\mathbf{x}_{n+1}$, ε_{n+1} is the associated error.

Eq. (5) presents a generalized expression for prediction bounds to

ensure broader applicability. The prediction bounds, representing the lower and upper limits of the PIs for new observations of the fitted models, can be expressed as:

$$y \pm t\sqrt{MSE + \mathbf{x}S\mathbf{x}^T}, \quad (5)$$

where y is the fitted value, MSE is the mean squared error, t depends on the confidence level, and is computed using the inverse of Student's t cumulative distribution function. S is the covariance matrix of the coefficient estimates, $\mathbf{x}$ is a row vector of the design matrix or Jacobian evaluated at a specified predictor value.

This process, although seemingly complex, can be efficiently carried out using various tools such as the Python libraries SciPy [35] and scikit-learn [36], as well as the MATLAB Curve Fitting Toolbox [37].

To evaluate the quality of PIs, four key indicators are commonly used: prediction interval coverage probability (PICP), mean prediction interval width (MPIW), normalized mean prediction interval width (NMPIW), and coverage width-based criterion (CWC). These metrics assess the reliability (PICP) and sharpness (MPIW and NMPIW) of the PIs, with CWC providing a comprehensive measure that balances both aspects [38].

PI coverage probability (PICP) of N observations is defined as

$$PICP = \frac{1}{N} \sum_{i=1}^N c_i, \quad (6)$$

where $c_i = 1$ if the predicted value $y_i \in [L(X_i), U(X_i)]$, otherwise $c_i = 0$. $L(X_i)$ is the lower bound of the i -th PI, $U(X_i)$ is the upper bound of the i -th PI. A PICP much lower than its nominal confidence value indicates that the constructed PIs lack reliability. If all predicted values fall within the range of all PIs, the PICP is 100%.

Wider prediction intervals fail to convey any information about variations in the target values. Therefore, the MPIW is adopted to quantify the width of PIs:

$$MPIW = \frac{1}{N} \sum_{i=1}^N (U(X_i) - L(X_i)). \quad (7)$$

MPIW can be normalized for objective comparison of PIs if the width of target (R) is known, normalized MPIW (NMPIW) can be given by Eq. (8):

$$NMPIW = \frac{MPIW}{R}. \quad (8)$$

The PICP measures the proportion of actual samples that fall within the predicted intervals, reflecting the reliability of the prediction. The NMPIW quantifies the average width between the upper and lower bounds of the prediction intervals, representing the sharpness of the prediction. Generally, for a given confidence level, higher PICP and lower NMPIW indicate better prediction performance. Since PICP and NMPIW are contradictory metrics, the CWC, which balances both metrics, can be used as a more comprehensive evaluation. CWC is a negative indicator, with smaller values indicating better performance:

$$CWC = NMPIW(1 + \gamma(PICP)e^{-\eta(PICP-\mu)}), \quad (9)$$

where η is recommended to be set to a value between 50 and 100, $\gamma(PICP)$ is a step function dependent on PICP and μ :

$$\gamma(PICP) = \begin{cases} 0, & \text{if } PICP \geq \mu \\ 1, & \text{if } PICP < \mu \end{cases}, \quad (10)$$

μ is set equal to the PI nominal confidence (PINC).

The above metrics will be applied in Section 3 to evaluate the reliability of the prediction intervals generated by various models. Furthermore, these metrics will be used to compare the reliability of prediction intervals across different models, offering insights for informed model selection [39].

2.3. Step 2: Typical condition identification

2.3.1. Definition of cooling tower working conditions

This step involves defining the operating conditions based on the input variables of the models to be fitted, as summarized in Table 2. This approach ensures that, once the input variables are specified, the model's output corresponds to a unique performance value.

For example, consider the York model outlined in Table 2. The equation for this model has input variables on the right-hand side, including outdoor wet-bulb temperature (T_{wb}), range temperature (T_{range}), and liquid-to-gas ratio (R_{lg}). The left-hand side represents the performance indicator, approach temperature (T_{app}). Thus, for the York model, a specific set of input variables, $([T_{wb}, T_{range}, R_{lg}])$, determines a corresponding output, (T_{app}) . Historical data can be extracted in the form $([T_{wb}, T_{range}, R_{lg}, T_{app}])$ for analysis, where (T_{app}) is calculated using readings from the cooling tower outlet water temperature sensor and the outdoor wet-bulb thermometer.

The same process applies to other models that require fitting. The extracted operating condition data can then be organized into a table to facilitate subsequent analysis.

2.3.2. Operating condition analysis and typical condition identification

As outlined in Section 2.3.1, defining operating conditions for cooling towers involves a complex integration of external meteorological conditions, internal cooling demands, and the effects of water and airflow on cooling performance. Due to multiple cooling towers typically operating in a rotational manner throughout the year, the operational range not only spans a whole year of outdoor temperature variations but is also subject to frequent changes in water-to-air mass flow ratio, influenced by responsive control logic of fans and condenser water pumps. Given that performance under varying operating conditions cannot be directly compared, a methodological approach is essential for discretizing input variables and compiling a performance table. The process, as depicted in Fig. 2, involves the following steps:

Step 1: Initial data preparation

Based on the chosen model expressions and input features, appropriate discretization intervals are determined for each variable. This determination is based either on the precision limits of the in-system sensors or on the operational range of features as experienced by maintenance personnel. Features and the associated performance indicators are extracted to form an initial raw table of dimensions $h \times (l+1)$ as $[\mathbf{x}, \mathbf{y}^{[obs]}]$, where h represents the sample size. A column vector of ones (dimension $h \times 1$) is appended to the **raw table** to facilitate subsequent total condition count calculations.

Step 2: Feature discretization

Using the discretization intervals defined in Step 1, each element of $\mathbf{x}$ in the raw table is assigned a grid index, resulting in an intermediate table of dimensions $h \times (l+1)$ as $[\mathbf{g}, \mathbf{y}^{[obs]}]$. Similar to Step 1, a column vector of ones (dimension $h \times 1$) is appended to the **intermediate table**.

Step 3: Condition proportion calculation

Create a new, empty **results table**. As each row from the **intermediate table** is processed: 1) The corresponding grid index vector is checked against existing entries in the results table; 2) *If not present*, the row vector and its count are added to the **results table**. *If present*, the count value from the current row is added to the existing count in the **results table**. Upon completion, the **results table** contains an $m \times l$ matrix of grid indices $\mathbf{g}$ and a corresponding $m \times 1$ vector of counts $\mathbf{C}$, where m is the total number of identified condition categories. The proportion $m \times 1$ vector $\mathbf{p}$ is calculated as $\mathbf{p} = (\mathbf{C}/h) \times 100\%$,

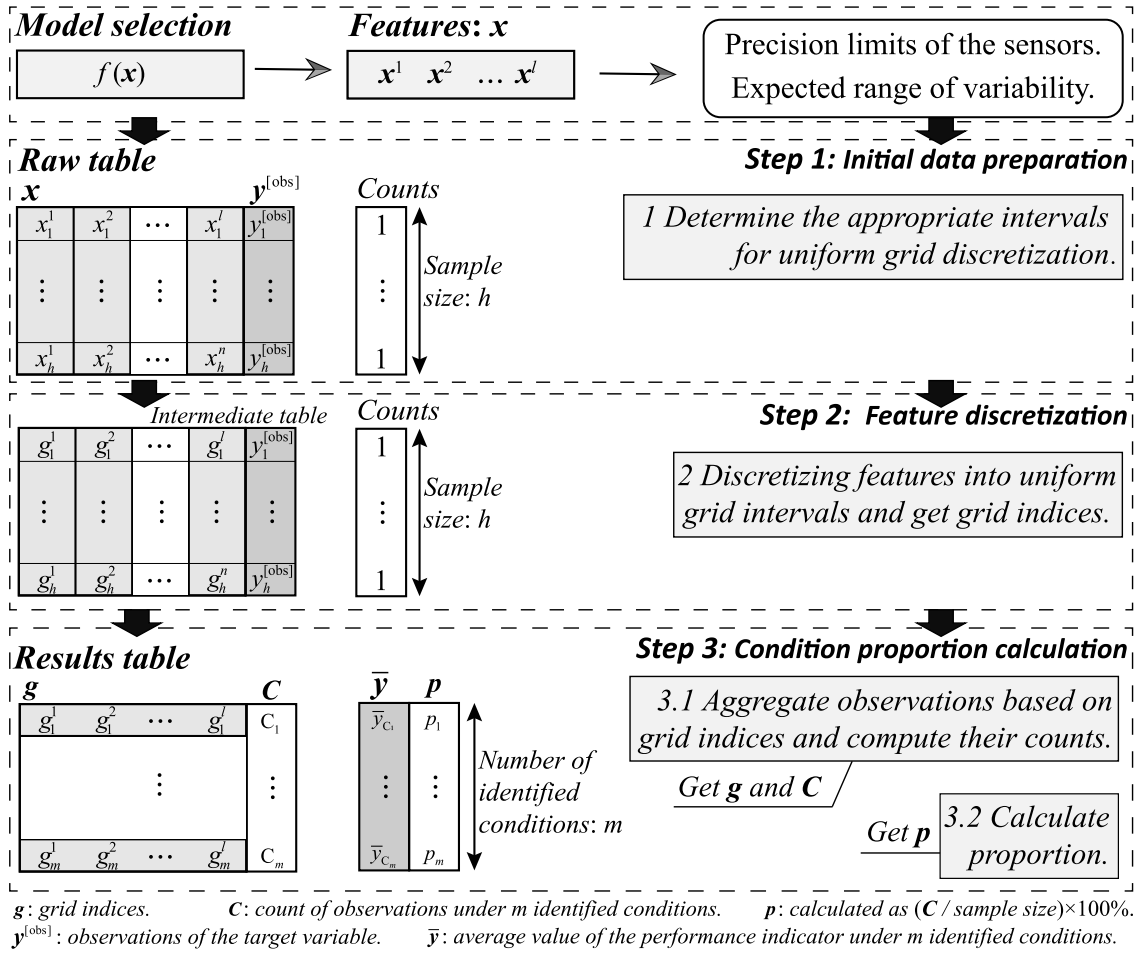


Fig. 2. The process of cooling tower operating condition analysis.

representing the percentage share of each condition. For each identified condition, the average value of the performance indicator is calculated using $y^{[obs]}$ from the **raw table**. This calculation yields an $m \times 1$ vector $\bar{y}$, which assists in subsequent performance evaluation analysis.

The results table, also referred to as the cooling tower thermal performance condition table, can facilitate the performance evaluation process based on the aggregated data insights.

2.4. Step 3: Performance evaluation

2.4.1. Evaluation metric selection

Commonly used evaluation metrics for assessing the thermal performance of cooling towers under specific operating conditions include the Merkel number, approach temperature, heat dissipation of the condenser water, and cooling tower capability. For the first three metrics, a higher Merkel number, lower approach temperature, and greater heat dissipation generally indicate higher cooling efficiency. For cooling tower capability, the fundamental approach is to evaluate the ratio of the actual cooling capacity to the design cooling capacity. This is typically calculated using the water flow rate handled under test conditions and the corresponding rating factor provided by the cooling technology institute (CTI). A capability ratio closer to or exceeding 100% indicates that the cooling tower's performance meets or exceeds the design requirements, reflecting superior performance under the tested operating conditions.

However, according to the CTI's *Acceptance Test Code for Cooling Towers* [31], applying cooling tower capability for thermal performance evaluation requires adherence to strict experimental protocols. Since this study is based on historical operational data rather than controlled

test conditions, the focus is placed on the first three metrics. Nonetheless, the methodology and references for implementing cooling tower capability as an evaluation metric are provided in [Appendix A](#) for additional context.

2.4.2. Equipment health assessment

The health assessment of cooling towers can assist maintenance personnel in detecting variations in thermal performance and implementing necessary corrective actions when performance falls below expected standards. In a real-world cooling system, thermal performance analysis should be conducted from two key perspectives.

First, for performance evaluation, characteristic curves can be derived based on physical principles and historical operational data. These curves provide insights into the cooling tower's historical performance trends without requiring additional testing. Second, for predictive maintenance, a performance forecasting approach should be adopted using models with high predictive accuracy. This allows for estimating cooling tower performance under user-specified operating conditions while providing corresponding PIs, enabling proactive maintenance decisions.

Several methods can be used to evaluate operational efficiency and identify potential issues: 1) *Comparison with design specifications*: assesses whether cooling towers operate as intended by comparing actual performance with manufacturer-provided design data. This requires detailed equipment specifications. 2) *Pre- and post-maintenance comparison*: evaluates the impact of maintenance (e.g., cleaning or part replacement) by comparing performance before and after maintenance activities. This relies on historical operational data and maintenance records. 3) *Benchmarking against peers*: compares the thermal

performance of cooling towers within the same system, focusing on units with identical design parameters. This method requires only historical operational data, facilitating internal benchmarking to identify under-performing units without external data, and will be adopted in the following analysis.

The subsequent analysis approaches cooling tower assessment from both performance evaluation and predictive maintenance perspectives. First, without requiring a system shutdown, it evaluates performance differences among cooling towers within the cooling plant using only historical operational data and physical principles. Second, it employs the most accurate predictive model to forecast cooling tower performance, providing a reference for predictive maintenance and serving as a benchmark for future system operations.

3. Case study

3.1. Case introduction

The cooling towers studied in this paper are configured in a “multi-tower – multi-pump – multi-chiller” arrangement within the cooling plant of a data center in Shanghai, China. As outlined in the Introduction, this flexible system configuration facilitates systematic equipment rotation, allowing for scheduled maintenance while optimizing the energy-saving potential of the system. However, the operation of the cooling towers is manually configured by maintenance personnel based on experience. As a result, despite all cooling towers being identical in model, their performance varies due to differences in cumulative operating hours. To fully leverage the advantages of this flexible system configuration, it is necessary to analyze cooling tower performance based on operational data and establish a predictive maintenance framework.

The collected operational dataset spans from May 2022 to October 2022, with a sampling interval of 5 minutes. Fig. 3 presents a generalized schematic of the studied system configuration, where $n = 6$ in this case, and each set of cooling equipment shown in the figure is identical in model. Both the cooling towers and condenser water pumps are equipped with VFDs, and the condenser water bypass valve remains closed. The cooling towers are dual-cell cross-flow cooling towers, with their engineering data and thermal performance data provided in Table 4 and Table 5, respectively. More detailed thermal performance data can be found in [40,41]. Specifications for flow meters and temperature sensors are listed in Tables B1–B2 in the Appendix.

3.1.1. Key assumptions for cooling tower thermal performance analysis

Before commencing the analysis, several key assumptions are made:

- A1. The water loss due to evaporation during the cooling tower operation is negligible and compensated by the makeup water.
- A2. The specific heat capacities of water and air remain constant along the cooling tower.

Table 4

Engineering data of the S3E-1222-12Q-2 cooling tower from the manufacturer (BAC).

Model number	Nominal capacity	Motor (power)	Fan (air flow)
S3E-1222-12Q-2	1737 [Tonnage] About 6109 [kW]	100 [HP] About 74 [kW]	408500 [CFM] About 193 [m ³ /s]

Table 5

Thermal performance data of the S3E-1222-12Q-2 cooling tower.

$\dot{m}_w$ [kg/s]	T_{wb} [°C]	T_{app} [°C]	T_{cwr} [°C]	T_{cws} [°C]
197	30.33	1.67	38	32
197	9.5	4	19.5	13.5

A3. The mass transfer coefficients between water and air remain constant along the cooling tower.

A4. Heat exchange between the cooling tower walls and the external environment is negligible.

A5. The preprocessed data assumes that the cooling tower operates under steady-state conditions.

A6. The operation of the cooling tower fan follows the affinity law, and the air flow rate can be estimated based on fan speed as follows:

$$\dot{V}_{air} = \dot{V}_{nominal} \cdot w_{fan}, \quad (11)$$

where $\dot{V}_{air}$ is the actual air volume flow rate, $\dot{V}_{nominal}$ is the rated air volume flow rate, w_{fan} is the current speed ratio of the cooling tower fan (defined as the current speed divided by the rated speed).

A7. Water flow is evenly distributed across all cells of the cooling tower during the same operating condition.

A8. The condensing water supply temperature during the same operation is calculated as the average of the outlet temperature readings from multiple cells of the cooling tower.

3.2. Model identification and PI evaluation

As discussed at the end of Section 2.2.2, three models were selected for fitting: the York model, Jin's model, and Zhou's model, with heat dissipation serving as the predicted variable (although the output of the York model is the approach temperature, heat dissipation was chosen as the prediction target to maintain consistency with the variables predicted by the other two models). The stochastic model identification was performed, and the model with the highest predictive accuracy and quality of prediction interval will be selected for further analysis on cooling tower thermal performance.

Fig. 4 shows the predictive accuracy for cooling tower heat dissipation. As observed, Jin's model exhibits the least favorable performance

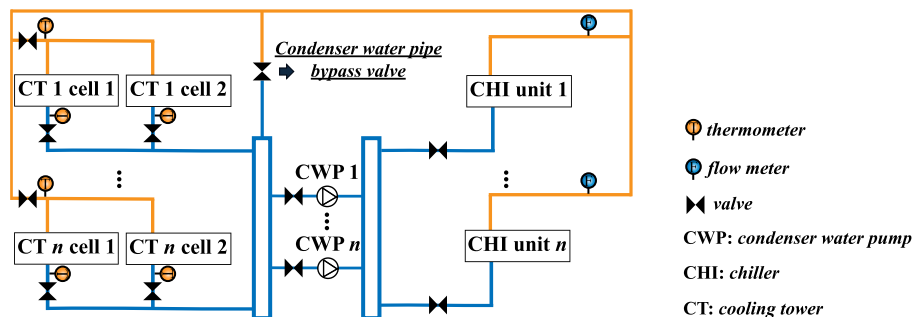


Fig. 3. Schematic of the condenser water loop with a multiple chiller and cooling tower configuration. (Abbreviations: CT – cooling tower, CWP – condenser water pump, CHI – chiller. In this study, $n = 6$).

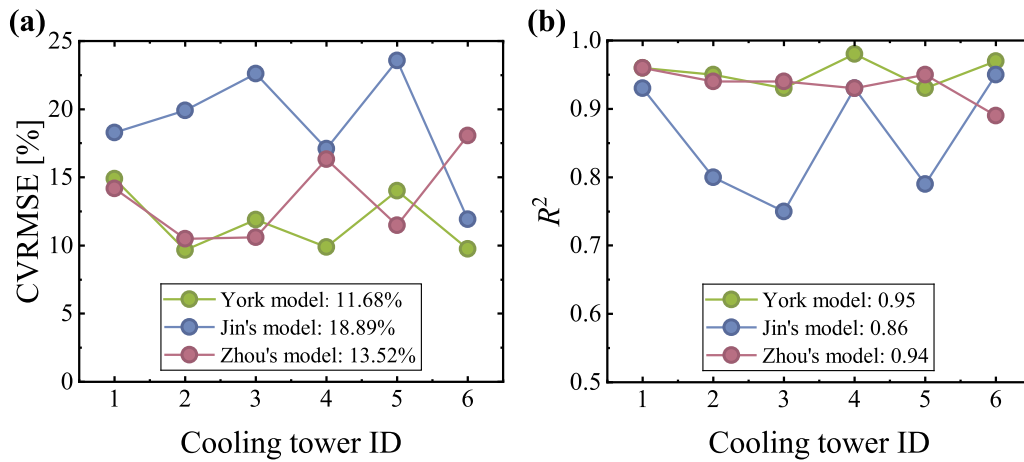


Fig. 4. Predictive accuracy for heat dissipation: (a) Coefficient of variation of the root mean square error (CVRMSE) across different cooling towers; (b) R-squared values across different cooling towers. (Average CVRMSE and R-squared values for each model are summarized in the legend.).

within the dataset, with a higher average CVRMSE of 18.89%, and a lower average R-squared value of 0.86. Detailed results can be found in Table B3 in the Appendix. Tables 6–8 present the quality of the constructed PIs for the three models, detailing PICP, NMPIW, and CWC values at nominal confidence levels of 90%, 80%, and 70%. As discussed in Section 2.2.3, a high-quality prediction interval must balance reliability with precision, aiming to achieve PICP values as close to or exceeding the nominal confidence levels while maintaining a smaller NMPIW (indicating narrower prediction intervals) to attain lower CWC values. Regarding the quality of PIs constructed by the three models: Firstly, in terms of reliability, at the 90% PINC level, the PICP values for the York model's prediction of heat dissipation across six cooling towers range from 87.35% to 90.27%. Jin's model shows a range from 82.2% to 91.76%, and Zhou's model from 84.03% to 89.25%. All three models generally exhibit PICP values below the PINC, with the York model's lower bound closest to 90%. At the 80% PINC, the York model's PICP values are above the PINC for all six cooling towers; Zhou's model ranges from 78.21% to 82.08%, close to or above the PINC; while Jin's model ranges from 75% to 83.03%, slightly underperforming in reliability compared to the York and Zhou models. At the 70% PINC, both the York and Zhou models' PICP values exceed the PINC, with Jin's model also approaching the PINC. This indicates that the York model performs best in terms of reliability, followed by Zhou's model, with Jin's model being the least reliable. Secondly, in terms of sharpness, a

Table 7

Quality of constructed PIs and model prediction accuracy (Jin's model, η in Eq. (9) is set to 50).

PI nominal confidence (μ)	Dataset	PICP [%]	NMPIW	CWC
90%	CT1	87.42	0.11	0.49
	CT2	83.43	0.13	3.66
	CT3	91.76	0.21	0.21
	CT4	83.97	0.09	1.90
	CT5	82.20	0.08	4.00
	CT6	88.57	0.09	0.26
80%	CT1	82.02	0.08	0.08
	CT2	75.00	0.10	1.36
	CT3	81.89	0.16	0.16
	CT4	78.17	0.07	0.24
	CT5	75.76	0.06	0.58
	CT6	83.03	0.07	0.07
70%	CT1	74.71	0.07	0.07
	CT2	68.52	0.08	0.26
	CT3	72.34	0.13	0.13
	CT4	70.76	0.06	0.06
	CT5	68.33	0.05	0.17
	CT6	73.19	0.05	0.05

Table 8

Quality of constructed PIs and model prediction accuracy (Zhou's model, η in Eq. (9) is set to 50).

PI nominal confidence (μ)	Dataset	PICP [%]	NMPIW	CWC
90%	CT1	89.25	0.09	0.23
	CT2	89.03	0.07	0.20
	CT3	87.60	0.07	0.31
	CT4	84.03	0.07	1.40
	CT5	88.01	0.07	0.26
	CT6	87.61	0.06	0.25
80%	CT1	82.08	0.07	0.07
	CT2	81.30	0.06	0.06
	CT3	79.85	0.06	0.11
	CT4	78.21	0.05	0.18
	CT5	81.36	0.05	0.05
	CT6	78.75	0.05	0.13
70%	CT1	74.15	0.06	0.06
	CT2	71.82	0.05	0.05
	CT3	72.23	0.04	0.04
	CT4	71.23	0.04	0.04
	CT5	73.33	0.04	0.04
	CT6	70.61	0.04	0.04

Table 6

Quality of constructed PIs and model prediction accuracy (York model, η in Eq. (9) is set to 50).

PI nominal confidence (μ)	Dataset	PICP [%]	NMPIW	CWC
90%	CT1	88.16	0.15	0.54
	CT2	90.27	0.15	0.15
	CT3	88.88	0.16	0.45
	CT4	87.35	0.13	0.64
	CT5	88.40	0.15	0.47
	CT6	88.29	0.18	0.61
80%	CT1	80.34	0.12	0.12
	CT2	81.39	0.12	0.12
	CT3	80.78	0.13	0.13
	CT4	80.17	0.10	0.10
	CT5	81.27	0.11	0.11
	CT6	81.57	0.14	0.14
70%	CT1	72.32	0.10	0.10
	CT2	71.20	0.10	0.10
	CT3	72.91	0.10	0.10
	CT4	72.12	0.08	0.08
	CT5	72.62	0.09	0.09
	CT6	73.20	0.11	0.11

comparison of the NMPIW values shows that all three models exhibit relatively low NMPIW values, with Zhou's model displaying the narrowest PIs, indicating higher sharpness than the other two models. Conclusively, when evaluating the balance between reliability and sharpness, ideal CWC values are low, indicating well-balanced models. In this context, Zhou's model and York model generally achieve lower CWC values compared to Jin's model, with Zhou's model exhibiting the best balance.

Based on the above analysis, the York model and Zhou's model will be utilized for further analysis of cooling tower thermal performance, specifically at the 80% nominal confidence level.

3.3. Typical condition identification and thermal performance assessment

This section analyzes the thermal performance of cooling towers from two perspectives. First, based solely on physical information and historical operational data, the thermal performance of the six cooling towers under the most frequently occurring typical operating conditions is evaluated (Section 3.3.1). This approach allows for identifying underperforming cooling towers without requiring system shutdown for testing. Second, based on two selected performance prediction models, the thermal performance of cooling towers is predicted under the targeted operating conditions, along with PIs. This provides a feasible framework for predictive maintenance (Section 3.3.2).

3.3.1. Analysis based on field data

According to the standards set by the CTI, a cooling tower characteristic curve represents a series of Merkel number values plotted under a specific operating condition while only the water-to-air mass flow ratio is adjusted. This characteristic curve typically demonstrates a linear trend on a log-log scale [42]. However, these curves are not usually included in the specifications provided by manufacturers. To derive the characteristic curve, one method is to perform a cooling tower test adhering to CTI standards. However, this approach is often impractical for operational systems due to the necessity for strict control over test conditions. An alternative approach is to analyze field data, identifying instances under specific operating conditions where the water-to-air mass flow ratio varies and subsequently calculating and comparing the corresponding Merkel numbers. In this case study, both the cooling towers and condenser water pumps are equipped with VFDs and operate year-round. Moreover, given that the system is located in Shanghai, where outdoor wet-bulb temperatures vary across a range of approximately 30°C, both the water-to-air mass flow ratio and ambient conditions demonstrate continuous fluctuations. Consequently, the collected field data is extensive, reflecting complex and highly dynamic operating conditions. Similar challenges are likely to be encountered in other real-world systems.

To address this, we adopt the procedure outlined in Section 2.3, leveraging the physical characteristics of cooling tower operation to define the characteristic curve using the selected input features: $[h_{sa}^{w,i} - h_{amb}, T_{w,i} - T_{wb}, \dot{m}_w/\dot{m}_a]$, where $(h_{sa}^{w,i} - h_{amb})$ represents the specific enthalpy difference between the inlet water and ambient air, $(T_{w,i} - T_{wb})$ denotes the temperature difference between the inlet water and the ambient air wet-bulb temperature, and $(\dot{m}_w/\dot{m}_a)$ is the water-to-air (dry air) mass flow ratio. These three variables serve as the defining conditions for the heat exchange process within the cooling tower, ensuring that the characteristic curve accurately reflects cooling tower thermal performance. This approach aligns with the third formulation of Zhou's model, which, with minor transformations, can be expressed as: $(T_{w,i} - T_{w,o}) = \psi[(T_{w,i} - T_{wb}), (h_{sa}^{w,i} - h_{amb}), (\dot{v}_w/\dot{v}_a)]$, where the left-hand side of the equation represents the heat exchange outcome, and ψ is a function that characterizes the relationship between these parameters.

For grid interval selection, the system configuration and operational characteristics were considered. In this case, the intervals of

$(h_{sa}^{w,i} - h_{amb})$ and $(T_{w,i} - T_{wb})$ were set as 5 [kJ/kg] and 2°C, respectively. These settings meet the requirements of subsequent analyses. When applying this method to other systems, the intervals can be adjusted based on system-specific operational characteristics. If higher precision is required, smaller intervals can be defined accordingly.

After processing the field data through statistical aggregation, a total of 58 unique combinations of specific enthalpy difference and temperature difference were identified. Table 9 summarizes the top five most frequently occurring operating conditions based on their proportions within the dataset. To comprehensively analyze the thermal performance of the six cooling towers, it is essential to select an operating condition from historical data where all six towers have enough data points. This ensures the reliability of the performance curve fitting, as an insufficient number of data points for a particular tower could lead to unreliable results. In this case study, the Condition #4 from Table 9 was selected for analysis.

Fig. 5 shows the cooling tower characteristic curve (Merkel number as a function of water-to-air mass flow ratio). As observed, the characteristic curve exhibits the expected linear trend on a log-log scale. For each cooling tower, a characteristic curve was fitted, with the red fitted line shown in each subfigure. A comparison of the performance differences among the six cooling towers reveals that Cooling Tower 1 exhibits significantly lower performance than the others, indicating a need for maintenance.

The scattered data points deviating from the fitted characteristic curve may be attributed to unstable operating conditions at certain sampling moments. This is due to the response delay of temperature sensors in the water system, as these sensors have a time constant (e.g., the Siemens QAE21 series temperature sensor used in this study has a time constant of approximately 30 seconds, see Table B2 in Appendix B) and do not immediately reflect adjustments in water flow and airflow. These discrepancies highlight the necessity of incorporating uncertainty quantification and prediction interval analysis, as will be discussed in the following section.

3.3.2. Analysis based on the two selected performance prediction model

As discussed in Section 3.2, this analysis focuses on the York model and Zhou's model at an 80% PINC level, specifically predicting approach temperature and heat flow. Compared to the Merkel number, these two metrics are simpler to monitor and calculate, enhancing their practicality for real-time comparison with operational data. In Section 3.3.1, the analysis identified that Cooling Tower 1 (CT1) exhibits lower performance than the others. Since all six cooling towers are of the same model, this section treats CT1's operational data as representative of performance degradation. To assess this, predictions are generated for both CT1 and Cooling Tower 4 (CT4) using a model calibrated on CT4's operational data. The effectiveness of the proposed predictive maintenance framework is validated by examining whether CT1's actual performance frequently falls outside the PI of the CT 4-based model (indicated by a significantly lower PICP than the expected 80% PINC). If CT1 consistently underperforms relative to predictions—demonstrating a larger approach temperature or a lower heat flow—this confirms the framework's ability to identify performance degradation.

The characteristic curve provides an intuitive visualization of cooling tower performance and its associated PIs. Therefore, this section

Table 9

Most frequent operating conditions and their proportions in the field dataset.

Condition ID	$(h_{sa}^{w,i} - h_{amb})$ [kJ/kg]	$(T_{w,i} - T_{wb})$ [°C]	Proportion [%]
#1	[35, 40)	[6, 8)	15.40
#2	[30, 35)	[6, 8)	11.40
#3	[25, 30)	[4, 6)	9.37
#4	[30, 35)	[8, 10)	8.64
#5	[40, 45)	[8, 10)	8.17

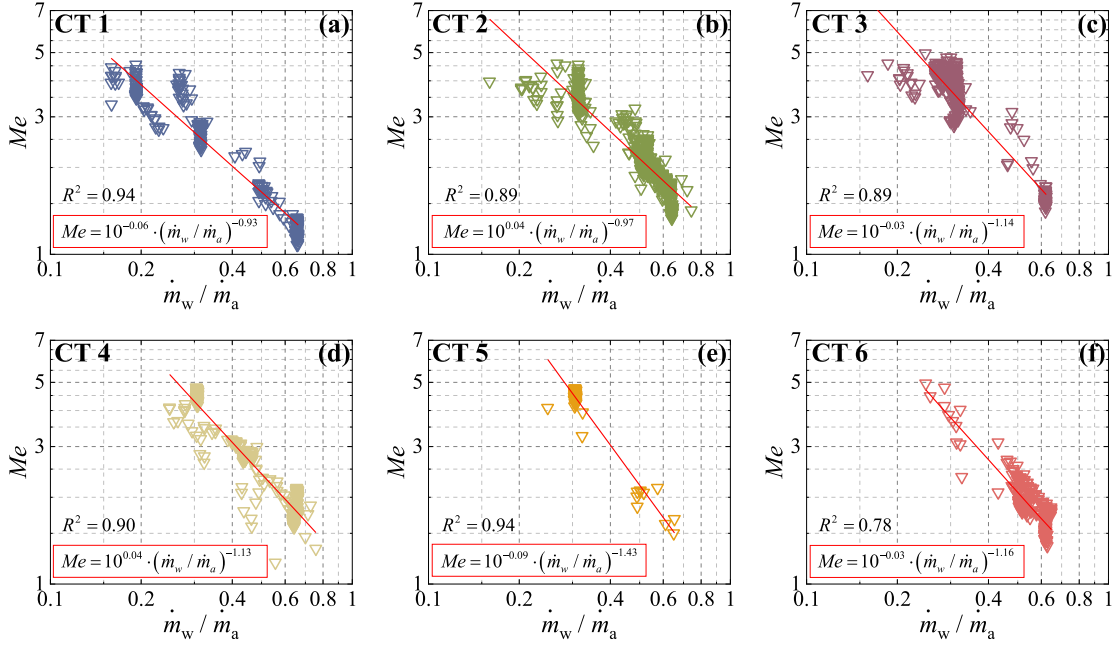


Fig. 5. Merkel number (Me) as a function of water-to-air (dry air) mass flow ratio ($\dot{m}_w/\dot{m}_a$). (CT refers to cooling tower; subfigures (a)–(f) correspond to cooling towers 1–6, respectively. The red line in each subfigure represents the fitted characteristic curve based on the data points.).

employs characteristic curves to illustrate the results of performance degradation analysis. To ensure sufficient field data for evaluating the performance of CT1 and CT4, the typical condition identification method proposed earlier is applied. Specifically, historical operating conditions for the York model and Zhou's model are clustered and ranked based on each model's input features. The ranking results are presented in Table 10 and Table 11, respectively, listing only conditions that account for more than 5% of all observed cases. For further analysis, operational conditions where both CT1 and CT4 have significant data points are selected. For the York model, Condition #2 in Table 10 is chosen, while for the Zhou's model, Condition #3 in Table 11 is used. To provide a comprehensive perspective:

- Ø For the York model, the characteristic curve is plotted with the wet-bulb temperature as the independent variable and approach temperature as the dependent variable.
- Ø For the Zhou's model, the characteristic curve is plotted with the water-to-air mass flow ratio as the independent variable and heat dissipation as the dependent variable. The water flow rate is adjusted proportionally to match the target water-to-air mass flow ratio, based on an assumed air volume flow of 115 m³/s within the selected air flow range [110,120) m³/s in Condition #3.

In Figures (6-7), the blue dashed line and shaded region represent the predicted characteristic curve and its corresponding PI. Specifically:

Table 10
Condition table based on York model.

Condition ID	T_{range} [°C]	R_{lg}	Proportion [%]
#1	[6, 8)	[0.4, 0.5)	9.67
#2	[4, 6)	[0.5, 0.6)	8.33
#3	[4, 6)	[1.2, 1.3)	7.21
#4	[4, 6)	[0.6, 0.7)	7.02
#5	[6, 8)	[0.5, 0.6)	5.85
#6	[6, 8)	[0.7, 0.8)	5.51
#7	[6, 8)	[0.6, 0.7)	5.01

Table 11
Condition table based on Zhou's model.

Condition ID	$(h_{sa}^{w,i} - h_{amb})$ [kJ/kg]	$(T_{w,i} - T_{wb})$ [°C]	$\dot{V}_{air}$ [m ³ /s]	Proportion [%]
#1	[35, 40)	[6, 8)	[110, 120)	8.34
#2	[30, 35)	[6, 8)	[110, 120)	6.64
#3	[30, 35)	[8, 10)	[110, 120)	5.85

Ø For the York model-based curve, predictions are generated using Table 10, Condition #2, assuming a T_{range} of 5°C and an R_{lg} of 0.55, in conjunction with the wet-bulb temperature on the x-axis.

Ø For the Zhou's model-based curve, predictions are generated using Table 11, Condition #3, assuming an enthalpy difference of 32.5 [kJ/kg], a temperature difference of 9°C, and an air volume flow rate of 115 m³/s. The water flow rate is proportionally adjusted to maintain the water-to-air mass flow ratio.

The trends in the predicted curves serve as a reference for assessing performance variations rather than precise predictive outcomes.

As expected, Figure 6(a) shows that the CT4-based model effectively predicts the approach temperature of CT4, achieving a PICP of 98% under an 80% PINC. In contrast, Figure 6(b) indicates that a significant portion of CT1's field data falls outside the PI of the CT4-based model, with a PICP of only 58%, and CT1 generally exhibits higher approach temperatures. This suggests that the CT4-based model is no longer a reliable predictor for CT1's performance, indicating thermal efficiency degradation. A similar pattern is observed in Fig. 7. In Figure 7(a), the CT4-based model accurately predicts CT4's heat flow rate, achieving a PICP of 96%. However, in Figure 7(b), when applied to CT1, the model yields a PICP of only 69%, with predicted heat flow rates generally exceeding CT1's actual values. These results are consistent with previous findings, further validating the proposed framework's ability to detect cooling tower performance deterioration.

Key considerations for Figs. 6 and 7:

1. Low approach temperatures in Fig. 6: The approach temperature in Fig. 6 appears low because the characteristic curve is plotted at an R_{lg} of 0.55. According to the manufacturer's performance specifications (Tables 4 and 5), this is reasonable. However, predictions below 0°C violate physical laws, highlighting the limitations of purely data-driven methods and the importance of incorporating physical constraints in performance evaluation.
2. Apparent discrepancy between Figs. 7 and 5 (Merkel model): At first glance, Fig. 7 appears inconsistent with Fig. 5 (Merkel model results in Section 3.3.1), where the Merkel number decreases with increasing x-axis values, while heat flow rate in Fig. 7 shows an increasing trend. However, this discrepancy can be explained by referring to the manufacturer's specifications (Tables 4 and 5). The observed increase in heat flow rate is due to the water flow rate remaining within operational limits. Additionally, in the logarithmic scale of Fig. 7, the increasing trend gradually flattens with higher water-to-air ratios, indicating that the cooling tower is approaching its maximum capacity. Beyond this threshold, heat flow rate ceases to increase significantly, as the growth in water flow surpasses the cooling tower's ability to further reduce temperature differential, leading to a diminishing rate of heat transfer increase.

4. Discussion

4.1. Model performance assessment

In this study, the Merkel model is used as the physical benchmark, and operational data from a real-world data center cooling plant is employed to fit three representative empirical and hybrid models: the York model, Jin's model, and Zhou's model. The evaluation of these models is conducted from three perspectives: fitting accuracy, practical applicability, and considerations for application in optimization-based control.

4.1.1. Fitting accuracy

From the fitting results, York model and Zhou's model exhibit better performance on the given dataset, achieving higher R^2 values and lower CVRMSE compared to Jin's model. For the PI quality, York model provides high-quality prediction intervals at 90%, 80%, and 70% PINC across all six cooling towers; Zhou's model performs well at 80% and 70% PINC but is less reliable at 90% PINC; Jin's model generates reliable prediction intervals only at 70% PINC. It is worth noting that in real-world applications, measurement uncertainty (such as sensor drift, delayed response, or calibration bias) can significantly influence model

fitting quality and prediction interval behavior. For instance, such uncertainty can widen the constructed PIs or reduce the PICP values, thereby affecting the reliability of degradation diagnostics. In this context, PICP and NMPIW metrics not only reflect the statistical performance of the model but also serve as indirect indicators of data quality. This highlights the importance of robust sensor validation and the careful interpretation of prediction metrics when deploying the framework in practice. Given these results, York model and Zhou's model are more suitable for cooling tower performance prediction in this study.

4.1.2. Practical applicability

York model has 27 fitting parameters and requires three input variables: range temperature, liquid-to-gas ratio, and outdoor wet-bulb temperature. Zhou's model has three fitting parameters and requires four input variables: enthalpy difference between inlet water and outdoor air, temperature difference between inlet water and outdoor air, water volume flow rate, and air volume flow rate.

York model's input variables are more straightforward to obtain, whereas Zhou's model requires enthalpy inputs, which need additional tools for calculation. In this study, CoolProp [43], an open-source thermophysical property library, was used to derive these values.

York model requires extensive data for parameter fitting due to the large number of adjustable parameters. However, its documentation provides a recommended set of default parameters that are applicable to most cooling towers, allowing users to skip the refitting process if these parameters sufficiently describe their system.

4.1.3. Considerations for application in optimization-based control

Though this aspect is not the primary focus of this study, it is worth a brief discussion. For models used in optimization applications, their mathematical properties need careful assessment:

York model's quadratic function structure: The monotonicity of the characteristic curves should align with physical principles. Certain parameter combinations may lead to unexpected turning points due to variations in parabolic shape and vertex location, which could affect the reliability of optimization outcomes.

Zhou's model's functional form: The fitted function must be evaluated for potential singularities within the relevant operational range.

These aspects highlight critical considerations for integrating these models into optimization frameworks. Improper handling of these factors can lead to unbounded solutions or physical inconsistencies, ultimately compromising the reliability of optimization-based control strategies.

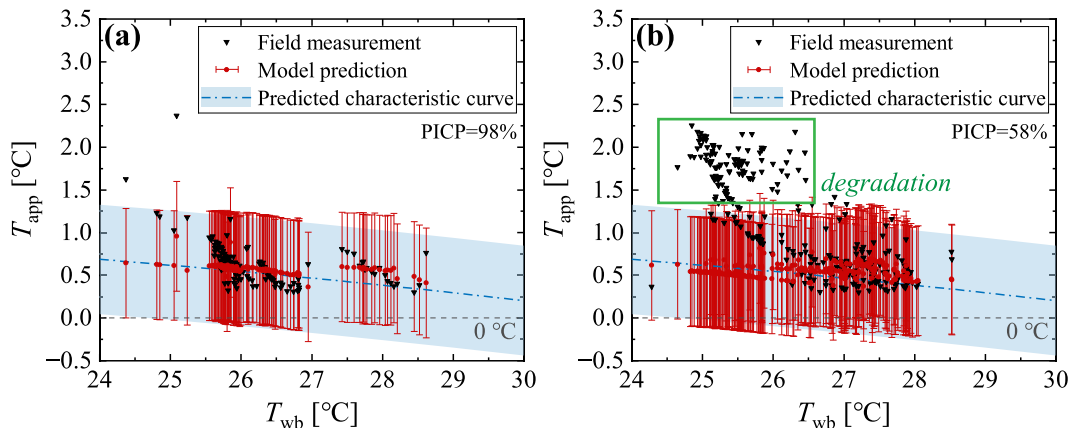


Fig. 6. Characteristic curve based on York model: (a) before performance degradation; (b) after performance degradation.

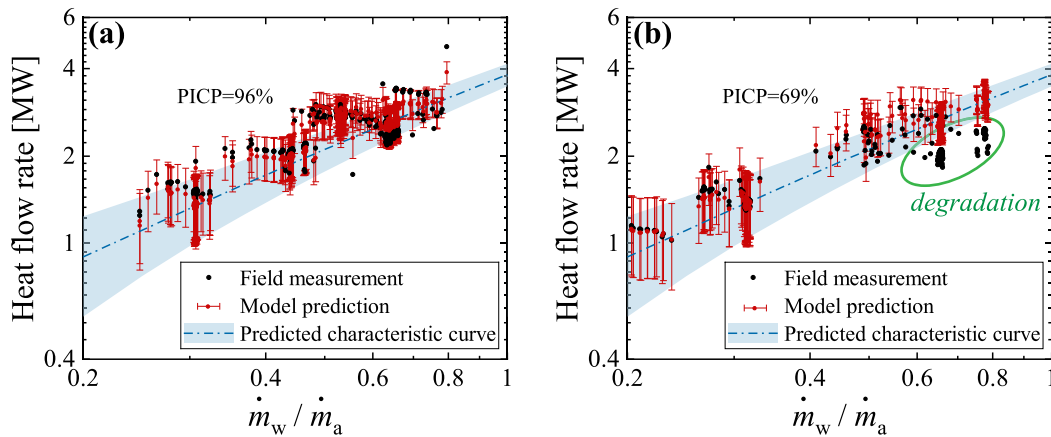


Fig. 7. Characteristic curve based on Zhou's model: (a) before performance degradation; (b) after performance degradation.

4.2. Predictive maintenance suggestions

As discussed in Section 3.1.1, this study is based on eight key assumptions (A1–A8). However, for cooling towers operating in real-world systems, A5–A8 require additional attention. Field data collected from operational cooling towers often exhibit the following characteristics:

1. Sensor accuracy and stability (A5 & A8): Temperature and humidity sensors may introduce measurement errors or have response delays due to their intrinsic time constants. This could result in data collection occurring before the system reaches a steady state.
2. Fan performance degradation (A6): Over time, fan performance deteriorates due to impeller wear, dust accumulation, or reduced transmission efficiency. This may cause actual airflow to deviate from values estimated using the affinity law, affecting thermal performance.
3. Uneven water distribution (A7): In actual operation, water flow across multiple tower cells may not be evenly distributed, deviating from the idealized assumption of uniform flow.

4.2.1. Diagnosing performance variations

When data analysis indicates performance degradation, it is crucial to systematically investigate the underlying causes:

1. Sudden performance decline: If cooling efficiency drops significantly over a short period, potential causes include fan transmission failures or unresponsive water valve adjustments.
2. Gradual performance degradation over time: If model predictions show a persistent PICP deviation from the expected PINC, maintenance personnel should inspect the cooling tower for: 1) Fouling accumulation, requiring cleaning; 2) Impeller wear, reducing airflow efficiency; 3) Fan efficiency loss, which can be assessed by comparing actual power consumption with the rated power.

4.2.2. Addressing sensor drift

Although random sensor errors are accounted for within the PIs, systematic sensor drift follows a different pattern and requires special attention. To detect potential drift: 1) Compare readings across multiple cooling tower units operating under similar conditions; 2) Utilize redundant sensor cross-validation techniques to identify anomalous temperature measurements; 3) Calibrate suspected drifted sensors promptly to maintain data accuracy.

4.3. Contributions

Compared to the existing evaluation methods provided by CTI, the approach proposed in this study enables the derivation of characteristic curves for cooling towers without requiring system shutdown, relying solely on historical operational data analysis. This method offers an efficient and practical means of comparing performance variations among identical cooling towers within a system, facilitating maintenance decision-making. In addition to its practical applicability, the proposed framework also provides theoretical contributions. By integrating physics-based insights (Merkel theory) with data-driven model fitting and prediction interval (PI) analysis, the framework ensures both physical interpretability and statistical rigor. The use of the deviation between the prediction interval coverage probability (PICP) and the nominal confidence level (PINC) as a performance degradation indicator establishes a model-agnostic and generalizable metric that can be applied across different cooling tower configurations and modeling approaches. Furthermore, this metric provides a theoretically grounded criterion that could be seamlessly integrated into future control or maintenance optimization frameworks. Furthermore, analysis using real-world data center operation records demonstrates that the proposed framework effectively manages large-scale operational data, extracts critical insights for performance evaluation, and accurately identifies performance differences and degradation among cooling towers. With high flexibility and scalability, this framework can be readily applied to various cooling systems, offering a practical solution for cooling tower performance assessment and predictive maintenance in engineering applications. The entire workflow of the proposed method is summarized in Fig. 8.

4.4. Limitations and future work

While the proposed framework demonstrates promising results using real-world operational data from a data center cooling plant, several limitations should be acknowledged. First, the framework has so far been validated only on a single site with relatively high-quality and high-frequency sensor data. Its applicability to other systems (particularly those with sparse data, low sensor resolution, or larger model inadequacies) requires further investigation. Second, although the framework accounts for typical measurement uncertainties, its robustness against extreme sensor drift or systemic errors remains to be tested. Third, the framework assumes that model structures are adequately chosen and calibrated; however, model selection in the presence of high uncertainty or limited domain knowledge may still be challenging. In future work, we aim to extend the evaluation of the framework to multiple buildings with varying configurations and data quality.

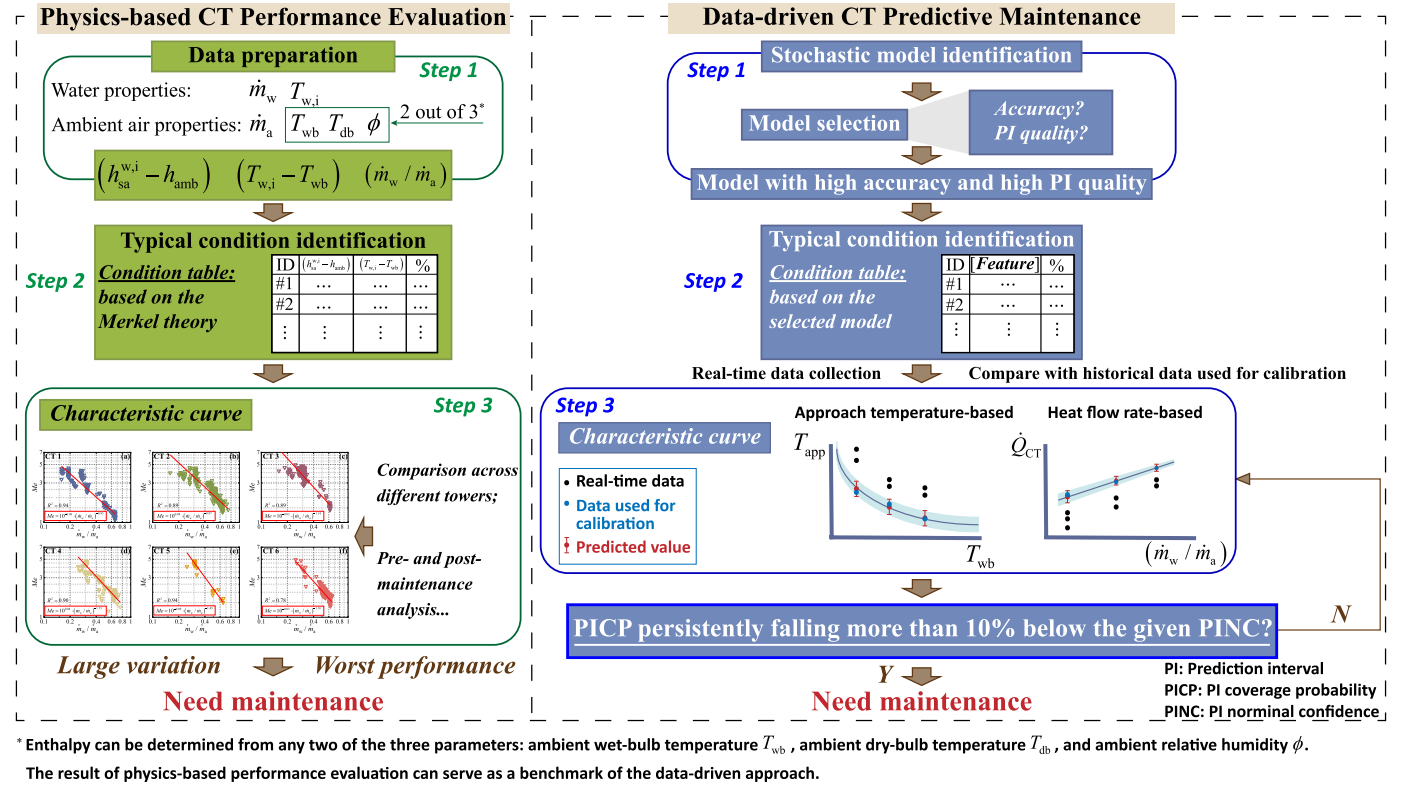


Fig. 8. The entire workflow of the proposed method.

5. Conclusions

This study proposes a practical performance evaluation framework toward predictive maintenance, enabling in situ assessment of cooling tower performance based solely on historical operational data, thereby eliminating the need for shutdown measurements. The framework integrates both physics-based and data-driven approaches and is validated using real-world operational data from a data center cooling plant, where Merkel theory serves as a benchmark for predictive model evaluations. Cooling tower thermal performance was assessed under typical operating conditions using an 80% prediction interval nominal confidence (PINC), and a prediction interval coverage probability (PICP) persistently falling more than 10% below the given PINC was identified as a potential indicator of performance degradation.

In the case study, the Merkel theory-based characteristic curve extraction method successfully identified CT1 as the least efficient cooling tower in the system. To further validate the predictive models' ability to detect performance deterioration, a model calibrated using CT4's data was applied to predict CT1's performance, and the results were compared against its field data. The predictive analysis employed the York model and Zhou's model, both of which demonstrated high predictive accuracy. The findings indicate that, at an 80% PINC: The York model for CT4 achieved a PICP of 98%, meaning 98% of CT4's actual data fell within the expected 80% confidence interval, demonstrating high prediction accuracy. However, when applied to CT1's data, the PICP dropped to 58%, with a significant number of data points exhibiting higher approach temperatures, suggesting that the CT4-based model can no longer reliably predict CT1's performance, indicating performance degradation. Similarly, the Zhou's model for CT4 yielded a PICP of 96%, while its PICP for CT1 was only 69%, with many data points showing lower heat flow rates, further reinforcing the observed performance decline.

Additionally, this study evaluated model performance from three

perspectives: fitting accuracy, practical applicability, and considerations for application in optimization-based control. Key factors for implementation and maintenance recommendations were discussed to support practical applications.

Compared to existing evaluation methods, the proposed method eliminates the need for complex lookup tables and interpolation, offering a streamlined and easily adaptable approach. While the current results are demonstrated in a single-case setting, they reveal a promising pathway for performance evaluation in multi-tower systems. With further validation across diverse scenarios, this framework can serve as a foundational tool for enabling energy-efficient and reliable predictive maintenance strategies.

CRedit authorship contribution statement

Si WU: Writing – original draft, Visualization, Software, Methodology, Investigation. **Pu YANG:** Resources. **Dingqian LI:** Resources. **Guanghao CHEN:** Writing – review & editing, Supervision, Funding acquisition. **Zhe WANG:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Reference to CTI test code for cooling tower thermal performance testing

For detailed guidelines on cooling tower thermal performance testing, refer to the CTI test code as cited in [31]. The relevant sections include:

Section VII: Thermal Performance Testing

E. Conducting the Test

F. Evaluation of Test Data

These sections provide comprehensive procedures for conducting performance tests and evaluating test data. For further details, see pages 99–102 of the referenced document.

Appendix B

Table B1

Technical data of flow sensors.

Sensor	Flow Transmitter DWM 2000
Current output	4 – 20 mA
Time Constant	5 seconds
Error Range	$v > 1\text{m/s}$, $\pm 2\%$ of the measured value (MV) $v < 1\text{m/s}$, $\pm 3\%$ of the measured value (MV)

Table B2

Technical data of temperature sensors.

Sensor	Siemens QAE21 series
Measuring range	$-30\dots+130^{\circ}\text{C}$
Time Constant	approx. 30 seconds
Error Range	$\pm 0.4 + 0.007 * \vartheta $ ϑ : Temperature in degrees Celsius

Table B3

Goodness of fit.

Model	Dataset	R^2	CVRMSE [%]
York model	CT1	0.96	14.90
	CT2	0.95	9.66
	CT3	0.93	11.89
	CT4	0.98	9.88
	CT5	0.93	14.01
	CT6	0.97	9.76
Jin's model	CT1	0.93	18.28
	CT2	0.80	19.90
	CT3	0.75	22.61
	CT4	0.93	17.09
	CT5	0.79	23.58
	CT6	0.95	11.92
Zhou's model	CT1	0.96	14.18
	CT2	0.94	10.49
	CT3	0.94	10.60
	CT4	0.93	16.33
	CT5	0.95	11.47
	CT6	0.89	18.07

(Table B1, Table B2, Table B3)

Data availability

Data will be made available on request.

References

- [1] W. He, J. Zhang, H. Li, S. Liu, Y. Wang, B. Lv, J. Wei, Optimal thermal management of server cooling system based cooling tower under different ambient temperatures, Appl. Therm. Eng. 207 (2022) 118176.

- [2] Y. Zhou, Q. Li, Z. Wang, S. Li, F. Wei, J. Liu, D. Yu, Thermal performance model of cooling towers for operational optimization: An equivalent temperature difference coefficient-based approach, *Appl. Therm. Eng.* 252 (2024) 123595.
- [3] N. Kumari Agarwal, P. Biswas, A. Shirke, Novel model predictive control by hypothetical stages to improve energy efficiency of industrial cooling tower, *Appl. Therm. Eng.* 215 (2022) 118899.
- [4] C.A.X. Marques, C.H. Fontes, M. Embiruçu, R.A. Kalid, Efficiency control in a commercial counter flow wet cooling tower, *Energ. Convers Manage.* 50 (2009) 2843–2855.
- [5] K. Ma, M. Liu, J. Zhang, Online optimization method of cooling water system based on the heat transfer model for cooling tower, *Energy* 231 (2021) 120896.
- [6] E. Al-Bassam, R. Alasser, Measurable energy savings of installing variable frequency drives for cooling towers' fans, compared to dual speed motors, *Energy Build* 67 (2013) 261–266.
- [7] Tutterow V, McKane AT. Var speed pump: guide success appl. 2004.
- [8] N. Trautman, A. Razban, J. Chen, Overall chilled water system energy consumption modeling and optimization, *Appl. Energy* 299 (2021) 117166.
- [9] H.W. Stanford III, HVAC water chillers and cooling towers: fundamentals, application, and operation, CRC press, 2011.
- [10] G. Wang, S. Ghoddousi, L. Ding, L. Song, Investigation of different cooling tower fan control strategies using COP of actual chillers and calibrated models of actual cooling towers and fans, *Energy Build* 277 (2022) 112585.
- [11] Z. Chen, J. Zhang, F. Xiao, K. Xu, Y. Chen, Development of a probabilistic cooling load prediction-based robust chiller sequencing strategy and its real-world implementation, *Appl. Energy* 382 (2025) 125213.
- [12] W. Zou, Y. Sun, Gao D-c, Z. Cui, Z. You, X. Ma, Robust enhancement of chiller sequencing control for tolerating sensor measurement uncertainties through controlling small-scale thermal energy storage, *Energy* 280 (2023) 128152.
- [13] S. Tang, D.R. Sheldon, C.M. Eastman, P. Pishdad-Bozorgi, X. Gao, A review of building information modeling (BIM) and the internet of things (IoT) devices integration: Present status and future trends, *Autom. Constr.* 101 (2019) 127–139.
- [14] K. Chen, X. Zhu, B. Anduv, X. Jin, Z. Du, Digital twins model and its updating method for heating, ventilation and air conditioning system using broad learning system algorithm, *Energy* 251 (2022) 124040.
- [15] Z. Shi, W. O'Brien, Development and implementation of automated fault detection and diagnostics for building systems: A review, *Autom. Constr.* 104 (2019) 215–229.
- [16] S. Wu, W. Zheng, Z. Wang, G. Chen, P. Yang, S. Yue, et al., AlphaDataCenterCooling: A virtual testbed for evaluating operational strategies in data center cooling plants, *Appl. Energy* 380 (2025) 125100.
- [17] X. Liang, X. Zhu, S. Chen, X. Jin, F. Xiao, Z. Du, Physics-constrained cooperative learning-based reference models for smart management of chillers considering extrapolation scenarios, *Appl. Energy* 349 (2023) 121642.
- [18] A.K. Aadithiyan, R. Sreeraj, B. Kiran Naik, S. Anbarasu, Assessment of evaporative cooling process across the mechanically driven cooling tower based on two-point boundary value problem using novel integral technique, *Int. J. Refrig.* 131 (2021) 254–262.
- [19] F. Merkel, Verdunstungskühlung, *VDI-Zeitschrift* 70 (1925) 123–128.
- [20] H. Jaber, R.L. Webb, Design of Cooling Towers by the Effectiveness-NTU Method, *J. Heat Transf.* 111 (1989) 837–843.
- [21] G-Y Jin, W-J Cai, L. Lu, E.L. Lee, A. Chiang, A simplified modeling of mechanical cooling tower for control and optimization of HVAC systems, *Energ Convers Manage* 48 (2007) 355–365.
- [22] M. Wetter, W. Zuo, T.S. Noudui, X. Pang, Modelica Buildings library, *J. Build Perform Simu.* 7 (2014) 253–270.
- [23] DOE-United States Department of Energy, Engineering Reference — EnergyPlus 23.2, Available at, <https://bigladdersoftware.com/epx/docs/23-2/engineering-reference/cooling-towers-and-evaporative-fluid-coolers.html#model-description-1-005>, 2023. Accessed: 03 April 2024.
- [24] M. Hosoz, H.M. Ertunc, H. Bulgurcu, Performance prediction of a cooling tower using artificial neural network, *Energ. Convers Manage.* 48 (2007) 1349–1359.
- [25] M. Gao, Sun F-z, Zhou S-j, Shi Y-t, Zhao Y-b, Wang N-h, Performance prediction of wet cooling tower using artificial neural network under cross-wind conditions, *Int. J. Therm. Sci.* 48 (2009) 583–589.
- [26] M. Gao, Shi Y-t, Wang N-n, Zhao Y-b, Sun F-z, Artificial neural network model research on effects of cross-wind to performance parameters of wet cooling tower based on level Froude number, *Appl. Therm. Eng.* 51 (2013) 1226–1234.
- [27] X. Qi, Y. Liu, Q. Guo, S. Yu, J. Yu, Performance prediction of a shower cooling tower using wavelet neural network, *Appl. Therm. Eng.* 108 (2016) 475–485.
- [28] H.H. Reichert, R.G. Donni, P.S. Schneider, ICA. Junior, Data driven assessment of a small scale evaporative condenser based on a combined artificial neural network with design of experiment approach, *Int. J. Refrig.* 115 (2020) 139–147.
- [29] M.C. Bueso, A.P. de Nicolás, F. Vera-García, Á. Molina-García, Cooling tower modeling based on machine learning approaches: Application to Zero Liquid Discharge in desalination processes, *Appl. Therm. Eng.* 242 (2024) 122522.
- [30] J.H. Kim, Y.S. Kim, H.G. Jo, E. Urabe, J. Mun, Y. Shin, et al., Accuracy, generalizability, and extrapolation ability of physics-based, data-driven, and hybrid models for real-life cooling towers, *Build Env* 274 (2025) 112756.
- [31] J.C. Hensley, Cooling tower fundamentals, Marley Cooling Tower Company Kansas City, 1985.
- [32] S. Wu, P. Yang, G. Chen, Z. Wang, Evaluating seasonal chiller performance using operational data, *Appl. Energy* 377 (2025) 124377.
- [33] J.C. Kloppers, D.G. Kröger, Cooling tower performance evaluation: Merkel, Poppe, and e-NTU methods of analysis, *J. Eng. Gas Turbines Power* 127 (2005) 1–7.
- [34] M. Poppe, H. Rögner, Berechnung von rückkühlwerken, *VDI-Wärmeatlas* 111 (1991) 1–15.
- [35] Virtanen P, Gommers R, Oliphant TE, Haberland M, Reddy T, Cournapeau D, et al. SciPy 1.0: Fundamental algorithms for scientific computing in python. *Nat. Methods*.17:261-272.
- [36] Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, et al. Scikit-learn: machine learning in python. *J. Mach. Learn. Res.*12:2825-2830.
- [37] The MathWorks Inc, MATLAB (R2023a), The MathWorks Inc., Natick, Massachusetts, 2023. <https://www.mathworks.com>.
- [38] A. Khosravi, S. Nahavandi, D. Creighton, A.F. Atiya, Lower Upper Bound Estimation Method for Construction of Neural Network-Based Prediction Intervals, *IEEE Trans. Neural Netw.* 22 (2011) 337–346.
- [39] B. Salimian Rizi, G. Pavlak, V. Cushing, M. Heidarinejad, Predicting uncertainty of a chiller plant power consumption using quantile random forest: A commercial building case study, *Energy* 283 (2023) 129112.
- [40] BAC, Series 3000 engineering data, Available at, https://baltimoreaircoil.com/download_api_endpoint/6026/C015_PAHBvi_3000_ED_20231020.pdf, 2023. Accessed: September, 2024.
- [41] BAC, Series 3000 (S3E/XES3E) cooling tower performance at standard conditions, Available at, https://baltimoreaircoil.com/download_api_endpoint/5015/S3000_ECTI_20210622.pdf, 2021. Accessed: September, 2024.
- [42] P. Navarro, J. Ruiz, M. Hernández, A.S. Kaiser, M. Lucas, Critical evaluation of the thermal performance analysis of a new cooling tower prototype, *Appl. Therm. Eng.* 213 (2022) 118719.
- [43] I.H. Bell, J. Wronski, S. Quoilin, V. Lemort, Pure and pseudo-pure fluid thermophysical property evaluation and the open-source thermophysical property library CoolProp, *Ind. Eng. Chem. Res.* 53 (2014) 2498–2508.